PRODUCTION RELEASE VALIDATION / 05 NOVEMBER 2026

TIO Operations Hub Production validation report

Production 2.53.0 | nb01994.vitronic.de

RELEASE SIGN-OFF: BLOCKED

Core workflows passed within the exercised sample. Outstanding defects and access gaps prevent full client-acceptance or security sign-off.

40	10 / 1 / 1	10 / 10
Settled routes observed	Automated pass / fail / blocked	Primary test entities cleaned

Three open defects

DEF-001 Missing HTTP-to-HTTPS redirect - Medium / P2

DEF-002 Incorrect empty-list record count - Low / P3

DEF-003 Structured audit metadata is unreadable - Medium / P2

Execution: 05 November 2026, 14:54:15-15:26:43 Asia/Dubai (32.5 minutes).

Test run: TIO-PROD-TEST-20261105-145415

All ten primary test entities were removed with approval and verified by authenticated 404 responses. Final live and ready checks passed. No production code deployment or schema change was performed.

This PDF preserves the findings of the completed validation run. It does not represent a new test execution. Detailed evidence filenames refer to the companion run folder.

Executive summary

Overall result: BLOCKED. Release recommendation: BLOCKED. Selected core workflows passed, but the available evidence does not support full client-acceptance or security sign-off. Three reproducible defects remain open: missing HTTP-to-HTTPS redirect, incorrect empty-list record count, and unreadable structured audit metadata. No critical defect was observed in the exercised workflows. These findings are not accepted risks, and a conditional release approval is not implied.

- Application: TIO Operations Hub **2.53.0**, Production, https://nb01994.vitronic.de/.
- Run: TIO-PROD-TEST-20261105-145415.
- Registered execution window: 05 Nov 2026 14:54:15-15:26:43 Asia/Dubai (32.5 minutes). The first browser discovery preceded run registration by approximately one minute.
- Identity: existing TIO Manager, explicitly approved by the user for isolated test records. Forms authentication, development sign-in disabled, all-client/all-internal scope, 244 rights observed.
- Automated unique checks: **10 passed, 1 failed, 1 blocked/skipped**, across the final suite and the additional responsive test. Earlier diagnostic reruns are retained and not counted again.
- Browser discovery: **40 settled routes**, plus isolated entity details. Module inventory and the precise scenario ledger are separate attachments.
- Cleanup: all **10 primary test entities** deleted after explicit action-time approval; each authenticated detail GET returned 404. Client cascade also removed its 15 secondary/history/quality/link rows (23 rows including the client); product and team were removed separately. Audit events remain intentionally.
- Production application code, schema, shared settings, real users and external integrations were not changed by this work. Existing unrelated local edits were preserved. No commit or deployment was made.

Coverage and interpretation

The table counts discrete checks in test-cases.csv. A route smoke test establishes an observed rendered page, not a completed business workflow. A compound unexecuted area is one coverage-gap item, not a claim to enumerate every possible test. Automated fixtures, source inspection, and observed production outcomes are distinguished. Only the recorded checks passed.

Module	Test cases	Passed	Failed	Blocked	Not tested
Archive	2	1	0	0	1
Audit	4	2	1	0	1
Clients	13	10	1	1	1
Commitments	9	8	0	0	1
Compatibility	2	1	0	1	0
Configuration	4	3	0	0	1
Contracts	8	7	0	0	1
Dashboard	4	3	0	0	1
Data quality	2	1	0	0	1
Environments	7	6	0	0	1
Escalations	2	1	0	0	1
Firewall rules	2	1	0	0	1
Handovers	2	1	0	0	1
Inventory	7	6	0	0	1
Jira integration	5	4	0	1	0
Notes	2	1	0	0	1
Notifications	2	1	0	0	1
Operations	3	1	0	2	0
Platform security	12	9	1	2	0
Products	5	4	0	0	1
Projects	4	3	0	0	1
Reassignment	2	1	0	0	1
Reporting	9	7	0	0	2
Roles and rights	4	3	0	1	0
Teams	4	3	0	0	1
Tickets	9	8	0	0	1
Users	3	1	0	1	1

The complete module inventory records URLs, purpose, available actions, observed access, dependencies, scenarios and risk: MODULE-INVENTORY.md (MODULE-INVENTORY.md). Per-check outcomes and evidence references are in test-cases.csv (test-cases.csv) and test-cases.json (test-cases.json).

Executed business workflows

Client and inventory: empty client required fields were rejected. Created client 22 as Opportunity; updated name/status; refreshed; verified duplicate and invalid-character rejection and a search returning one record. Created Test environment 10, updated its synthetic location/description, reported Healthy with a run-prefixed reason, then recorded metadata-only virtual server 86 and product installation 10. Parent overview subsequently showed one installation, one asset and Healthy condition. Unknown version/specification values stayed explicit. No real host was contacted or provisioned.

Project and contract: created Planned project 6 with synthetic product 5. Created Draft contract 7 with no payments, penalty clauses or Jira connection. The server rejected an end date before its start. A corrected 2027-2028 term and synthetic obligations persisted. Archived and restored the contract through the authorized archive UI; archive and restoration history remained. The contract never became Active.

Ticket: CS-2760 / 10960 belonged to client 22, synthetic team 3, the approved current owner and test environment 10. Executed New → triage (display Open) → In progress → Waiting for customer → In progress → Closed with synthetic outcome/evidence. The timeline retained six workflow events and one waiting interval. No customer response, external dispatch or Jira write was sent. No contractual SLA was linked or claimed measured.

Commitment: CM-1012 / 12 belonged to client 22 and the current owner. Creation, deadline revision, preserved original deadline, revision count, rejection of a future completion, and completion with evidence all passed. It displayed Completed on time; audit showed UTC deadline 2027-12-02T08:00:00Z corresponding to 12:00 Asia/Dubai. Native date-field keyboard events were needed after automation fill; the first create retained its default 12 Nov 2026 deadline. This was corrected through the audited revision and is recorded as an automation limitation, not an inferred application defect.

Deletion: reviewed exact dependent IDs before approval. Client cascade removed the isolated environment, asset, installation, project, Draft contract, closed ticket and completed commitment, together with generated dependent rows. Product and team were then deleted after their links had cleared. Every primary entity was read again with the same manager session and returned a 404 Problem Details response. Client search returned no matches. This verifies application-level cleanup; no direct SQL sweep was available.

Defect register

DEF-001 - HTTP root does not redirect to HTTPS

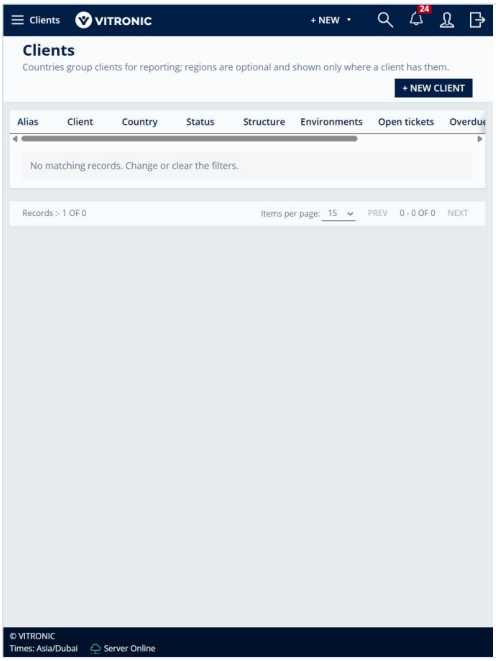
- Module: IIS / transport. **Severity: Medium. Priority: P2.** Open; no fix deployed.
- Reproduce: anonymous GET http://nb01994.vitronic.de/ with redirects disabled.
- Expected: 301/302/307/308 with an HTTPS Location header (or an explicitly accepted HTTPS-only refusal).
- Actual: 200, empty body, Content-Length 0, Microsoft-IIS/10.0, no Location. Reproduced in the main run and focused diagnostics. HTTPS itself worked and supplied HSTS.
- Impact: an initial HTTP entry does not reach the secure application. No HTTP login page or credential exposure was observed; that consequence is not asserted.
- Evidence: HTTP response and failure trace (automated-complete/playwright-results.json), automated-complete/artifacts/readonly.production.ts-HTTP-redirects-to-HTTPS/trace.zip, focused response (http-diagnostic/playwright-results.json). A screenshot is not applicable to this API-level empty response.
- Action: review IIS port-80 bindings and approved redirect policy, correct configuration under separate approval, then rerun the check. Server access logs were unavailable.

Observed HTTP response	Value
Status	200
Location	Absent
Server	Microsoft-IIS/10.0
Content-Length	0
Response body	Empty

Defect register

DEF-002 - Empty filtered client pager displays 1 OF 0

- Module: Clients / shared pagination. **Severity: Low. Priority: P3.** Open.
- Reproduce: open /clients?q=<a unique string with no matching clients>; the run-prefix search after cleanup provided this condition.
- Expected: Records :- 0 OF 0, range 0 - 0 OF 0; navigation disabled.
- Actual: Records :- **1 OF 0**, range **0 - 0 OF 0**; Prev/Next disabled.
- Impact: misleading record count in an empty result. No data loss observed.
- Evidence: screenshot (empty-client-pagination.png), browser-evidence.json#DEFECT-EMPTY-PAGER. Repository review found the shared Pager count expression in src/web/src/ui/kit.tsx; no code was changed.
- Action: correct the empty-range count and verify empty, single-page and multi-page lists. Server logs are not needed to reproduce this UI rendering issue.



Evidence: the empty result has range 0 - 0 OF 0, while the record count states 1 OF 0.

Defect register

DEF-003 - Structured audit metadata renders as empty before/after pairs

- Module: Change history. **Severity: Medium. Priority: P2.** Open.
- Reproduce: /admin/audit, Record type Client, Action Remove; inspect the row labelled TIO-PROD-TEST-20261105-145415.
- Expected: readable named deletion counts (for example Environments 1, Servers 1, Tickets 1) and unlink metadata.
- Actual UI: deleted: ∅ → ∅, unlinked: ∅ → ∅. Existing export events also displayed truncated scalar values, such as format: X → l.
- API evidence: the same run-owned removal event retains the complete deleted object and empty unlinked object. **The stored audit data was not lost.**
- Impact: privileged-operation investigation is obscured in the UI, including the cleanup event from this run.
- Evidence: raw run-owned audit event (own-client-deletion-audit.json), screenshot (client-delete-audit.png), browser-evidence.json#DEFECT-AUDIT-METADATA. Repository Changes renderer assumes every JSON value is a two-element array in src/web/src/pages/common.tsx.
- Action: render structured metadata and scalars according to their shape while preserving ordinary before/after pairs; verify cascade and export events. Server application logs were not accessible.

Stored API data versus observed UI text

Field	Stored API metadata	Observed UI display
deleted	Environments: 1; Servers: 1; Installations: 1; Projects: 1; Contracts: 1; Tickets: 1; Commitments: 1; plus 15 secondary rows.	∅ → ∅
unlinked	Empty object: {}	∅ → ∅

Source: own-client-deletion-audit.json and browser-evidence.json#CLIENT-DELETE-AUDIT. The screenshot referenced in the source report captures the retained event; its Changes column is outside the captured horizontal viewport.

Operational findings

Availability and performance. Before testing, readiness returned 200. Final live and ready bodies were {"status":"live"} and {"status":"ready"}, and the authenticated dashboard rendered after cleanup. Final automated liveness/readiness test durations were 69/26 ms; anonymous protected-resource test durations were 8-12 ms; fresh Edge sign-in test took about 1.4 s and responsive keyboard test about 1.8 s. These are low-volume, client-side test durations, not server p95/p99 or controlled application performance measurements. No load, stress or availability-SLA claim is made. Early browser-navigation timings captured loading states and were excluded.

Security. Five anonymous protected resources returned 401: me, clients, tickets, admin users and OpenAPI. A fresh Edge browser required sign-in. HTTPS responses had HSTS, CSP with frame-ancestors none, X-Frame-Options DENY, nosniff, Referrer-Policy and restrictive Permissions-Policy; API health responses used no-store. HTTPS antiforgery cookies were Secure/SameSite Strict, with HttpOnly on the server antiforgery cookie and a deliberately readable XSRF request-token cookie. Cookie values were redacted. Authentication-cookie attributes, logout invalidation, expiry, and cross-role/object-scope enforcement were not established. No credential attacks, authorization bypass or intrusive payloads were used. Anonymous responses did not reveal connection strings, password hashes or stack traces under the tested assertions.

Integration. The deployed OpenAPI describes 273 paths and a read-only Jira integration. Connections, review, conflicts and monitor screens loaded; recent scheduled synchronization rows showed Success. This is historical telemetry, not an actively triggered connectivity, retry or timeout test. Existing Jira integration issues were visible in audit activity and require mapping/operations review; they were not changed. No sync, credential rotation, discovery, recalculation or third-party call was manually triggered.

Data integrity. New records, updates, relationships, UTC/local display, audit history, duplicate protection, archive/restore and deletion were observed through the application. Database readiness proves its configured readiness conditions, not a full referential-integrity audit. No database credentials, SQL session, direct orphan queries, transaction-failure injection or schema inspection were available. Audit records intentionally persist after entity removal.

Observability. In-app audit and Jira run history were accessible. No error/warning console messages were returned for the inspected signed-in tabs before cleanup. The fresh Edge sign-in test saw no page errors, failed requests or 5xx responses under its listeners. There was no comprehensive authenticated network trace, host/IIS logs, alerts, dashboards of infrastructure metrics, backup/restore drill or background-job telemetry for every worker. Expected negative-test errors and post-deletion 404s are not availability defects.

UI/compatibility. Desktop forms, dialogs, list search, empty states, field messages and workflow labels were exercised. A fresh Edge sign-in page passed a real 768×1024 viewport assertion and keyboard movement from account to password. The embedded browser viewport override was ineffective (actual DOM stayed 1280×720), so its image named ticket-tablet.png is **not tablet evidence**; authenticated responsive coverage remains blocked. Full accessibility/contrast/screen-reader and other browser-engine coverage were not executed.

Limits and release follow-up

The report does not certify the full application. Important unexecuted areas are explicit in the coverage ledger: alternate accounts and scopes; real sign-in/logout/expiry; all editable fields and concurrent UI saves; nested catalogue/contact/organization/firewall/attachment workflows; escalation and handover lifecycles; real contractual SLA/holiday/penalty/payment flows; report generation/download/schedules; notification delivery; all integration failure modes; host monitoring and SQL integrity. Shared configuration, access changes and uncertain recipient effects were left untouched. No external services or migrations were started.

Required follow-up: triage DEF-001 and DEF-003, fix DEF-002, supply approved alternate test accounts and a reproducible test session, arrange read-only host/monitoring/SQL evidence, and execute the remaining release-critical scenarios in an isolated UAT copy or expressly approved production resources. Recheck live/ready and the production suite after any approved correction. An owner must decide mandatory acceptance criteria and accept any non-blocking residual issues before a GO or CONDITIONAL GO can be supported.

Test data and database impact

All ten registered primary entities were removed after named approval; no primary test records require manual cleanup. The cascade preview counted 22 dependent rows plus the client. Generated secondary rows included a condition report, two data-quality items, project-product and ticket-environment links, one waiting interval and workflow/commitment history. The audit retains the exact counts. Immutable audit events are intentional retained evidence. Creation times in the registry are explicitly first-observation timestamps, not invented SQL CreatedAt values. No genuine client records, user accounts, shared policies, schema, migrations or database maintenance were changed by this run.

See test-data-registry.json (test-data-registry.json), cleanup-verification.json (cleanup-verification.json), cleanup dependencies (cleanup-dependencies.json), and post-cleanup-health.json (post-cleanup-health.json).

Primary test-data cleanup register

Entity	Record ID	Cleanup outcome
Client	22	Deleted; authenticated GET returned 404
Environment	10	Deleted; authenticated GET returned 404
Product	5	Deleted; authenticated GET returned 404
Commitment	12	Deleted; authenticated GET returned 404
Team	3	Deleted; authenticated GET returned 404
Project	6	Deleted; authenticated GET returned 404
Case	10960	Deleted; authenticated GET returned 404
Contract	7	Deleted; authenticated GET returned 404
Installation	10	Deleted; authenticated GET returned 404
Asset	86	Deleted; authenticated GET returned 404

Creation timestamps in the companion registry are first observations, not exact SQL CreatedAt values. Immutable audit events remain as evidence.

Automation and repository handoff

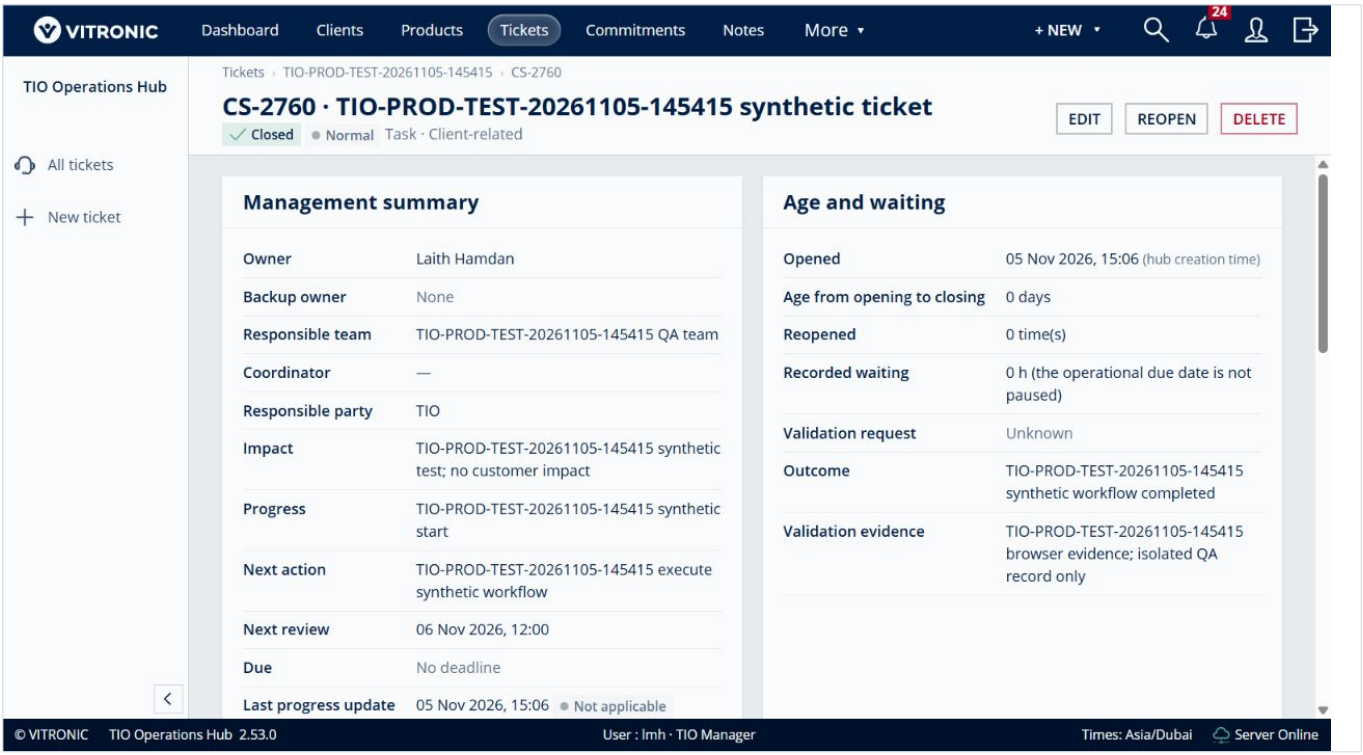
Added a separate production Playwright configuration, GET-only health/auth/header/cookie/sign-in checks, and a guarded authenticated client/commitment lifecycle suite with stale-update protection and ID-specific cleanup. The latter is **implemented but not production-executed** because no approved Playwright session file was supplied. It is skipped by default; authenticated traces are disabled to avoid storing session cookies. No default development config, source code or packages were changed. The .production.ts filenames prevent discovery by the ordinary development suite.

Run instructions are in tests/e2e/production/README.md. TLS verification remained enabled. Node 24 needed NODE_USE_SYSTEM_CA=1 for the organization CA; the first unconfigured run was interrupted during harness teardown, and is not treated as product test evidence. Machine-readable JSON/JUnit, a retained HTTP failure trace, DOM evidence, screenshots, test data and this report are saved in this run directory.

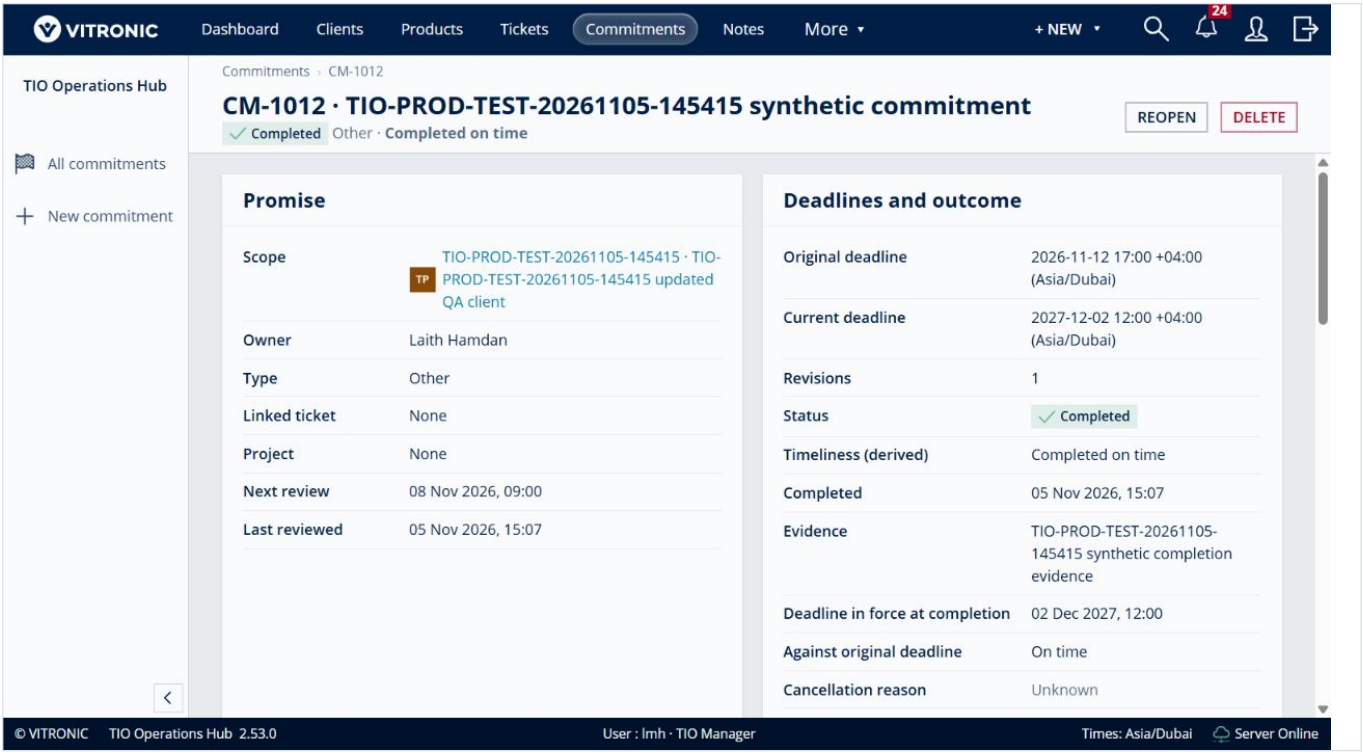
Executed checks: production Playwright main suite (9 pass, 1 fail, 1 skip), additional tablet/keyboard test (1 pass), focused HTTP/cookie/trust diagnostics, 40 settled-route observations, selected isolated workflows and cleanup verification. No .NET, frontend application build, default development Playwright suite, migrations or test servers were run: application code was unchanged and the development suites assume database setup. git diff --check was run for the new test directory. A separate TypeScript no-emit check failed because this checkout lacks installed Node type definitions (node:fs/promises, node:path, process); no dependencies were installed. Playwright successfully transpiled/executed the read-only suite and discovered the gated write suite. Its full static type check and approved UAT/production execution remain outstanding.

Files added: tests/e2e/production/{playwright.config.ts,readonly.production.ts,isolated.production.ts,README.md} and outputs/TIO-PROD-TEST-20261105-145415/ (report, inventory, ledgers, evidence and report generator). Existing dirty-tree changes were not committed, changed or reverted.

Appendix A - Workflow evidence

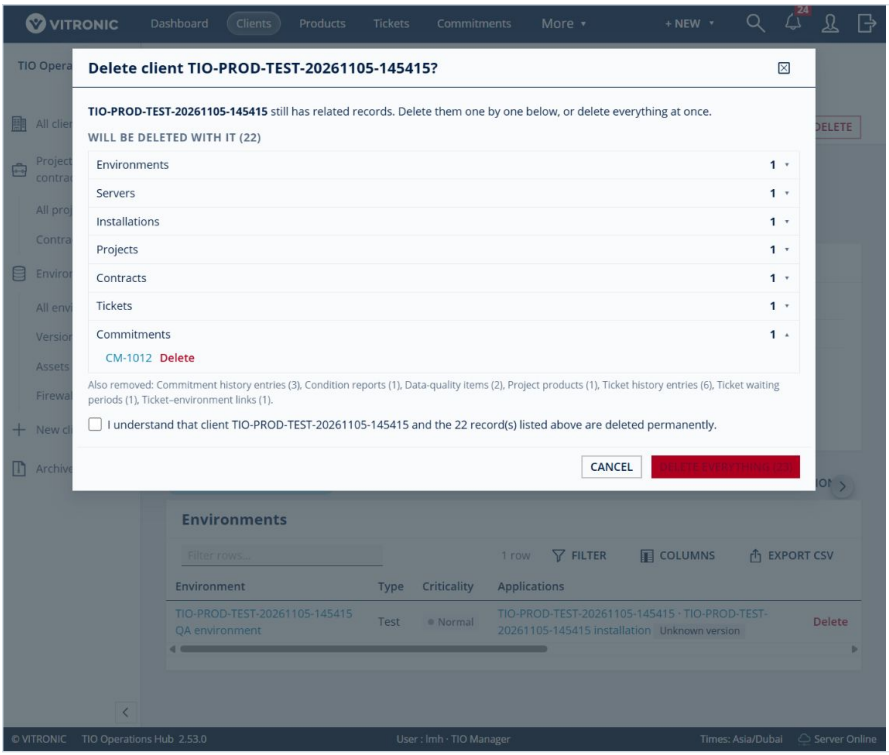


A1. Closed synthetic ticket CS-2760 with its test-owned team, outcome and validation evidence.

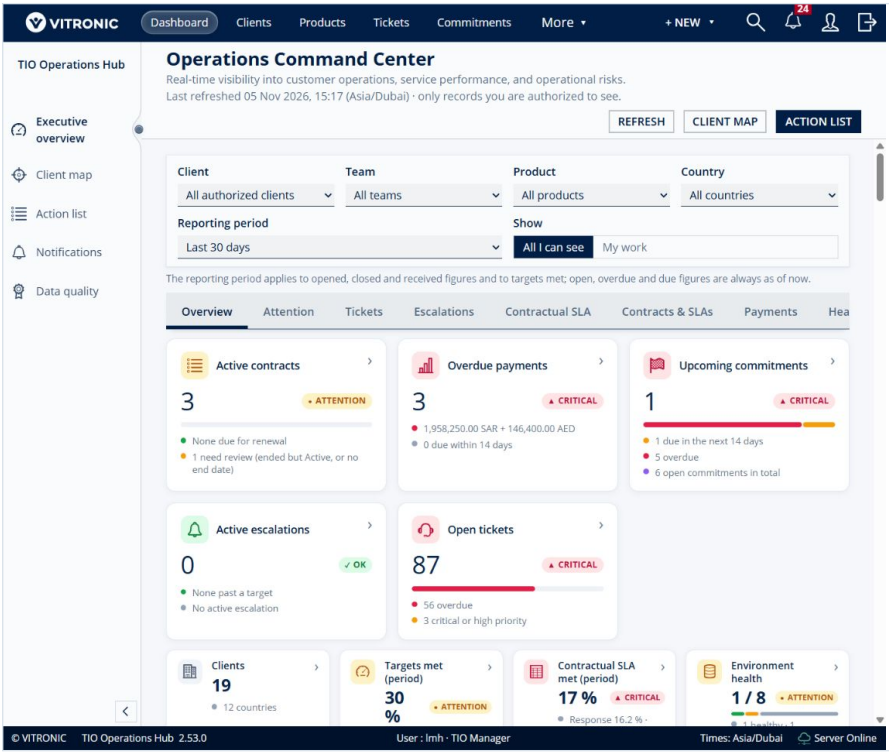


A2. Synthetic commitment CM-1012 after audited deadline revision and completion; displayed as completed on time.

Appendix A - Cleanup and health



A3. Reviewed dependency preview before approved permanent cleanup. The primary client owned all listed test records.



A4. Authenticated dashboard after cleanup. Final live and ready endpoint bodies were healthy.

Appendix B - Confirmed route index

The executive overview and 39 additional routes were observed in settled states. A rendered page is smoke evidence only; it does not establish that all controls or workflows passed. All observations used the approved TIO Manager account with all-client/all-internal scope.

Module / workspace	Confirmed URL path
Operations Command Center	/
Clients	/clients
Products	/products
Tickets	/cases
Commitments	/commitments
Notes workspace	/notes
Escalations	/escalations
Teams and coverage	/teams
Reporting Center	/reporting-center
Projects & contracts	/projects
Projects & contracts	/contracts
Environments	/environments
Product version matrix	/versions
Infrastructure assets	/assets
Firewall rules	/firewall-rules
Handovers	/handovers
Reassignment queue	/reassignment
Roles	/admin/roles
Rights	/admin/rights
Access review	/admin/access
Configuration	/admin/config
Review policies	/admin/policies
Jira Connections	/admin/integrations/jira
Jira import review	/admin/integrations/jira/review
Jira conflict resolution	/admin/integrations/jira/conflicts
Jira synchronization monitoring	/admin/integrations/jira/monitoring
Change history	/admin/audit
Archived records	/admin/archived
Action list	/actions
Notifications	/notifications
Data quality	/data-quality
Client map	/map
Settings	/admin/settings
Reporting: library	/reporting-center/library
Reporting: builder / cases	/reporting-center/builder/cases
Reporting: my-reports	/reporting-center/my-reports
Reporting: schedules	/reporting-center/schedules
Reporting: history	/reporting-center/history
Reporting: templates	/reporting-center/templates
Users	/admin/users

Dynamic test details were additionally exercised: client 22, environment 10, server 86, installation 10, project 6, contract 7, ticket 10960, commitment 12, product 5 and team 3. All were subsequently deleted. The deployed API contract exposed 273 paths; this is discovery, not complete endpoint coverage.